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7MA2

Unsharp Masking

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image = cv2.imread('Parrot.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB) # Convert from BGR to RGB

# Gaussian sigma
sigma=2

#parameter
amount=1.5

# Convert image to float32 for processing
image_float = image.astype(np.float32) / 255.0

# Apply Gaussian blur
blurred = cv2.GaussianBlur(image_float, (0, 0), sigmaX=sigma,
sigmaY=sigma)

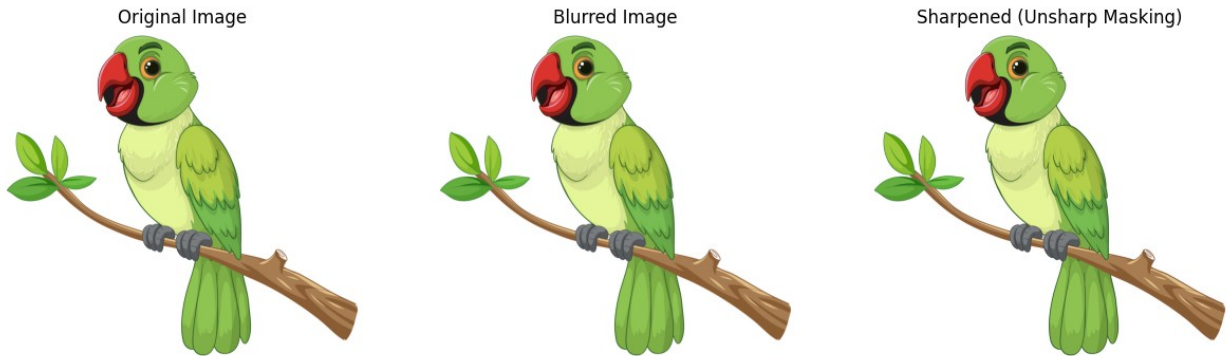
# Perform Unsharp Masking
sharpened = np.clip(image_float + amount * (image_float - blurred), 0,
1)
# Plot original, blurred, and sharpened images
fig, ax = plt.subplots(1, 3, figsize=(15, 5))

ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(blurred)
ax[1].set_title("Blurred Image")
ax[1].axis("off")
```

```
ax[2].imshow(sharpened)
ax[2].set_title("Sharpened (Unsharp Masking)")
ax[2].axis("off")

plt.show()
```



Task#1

```
image = cv2.imread('Parrot.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

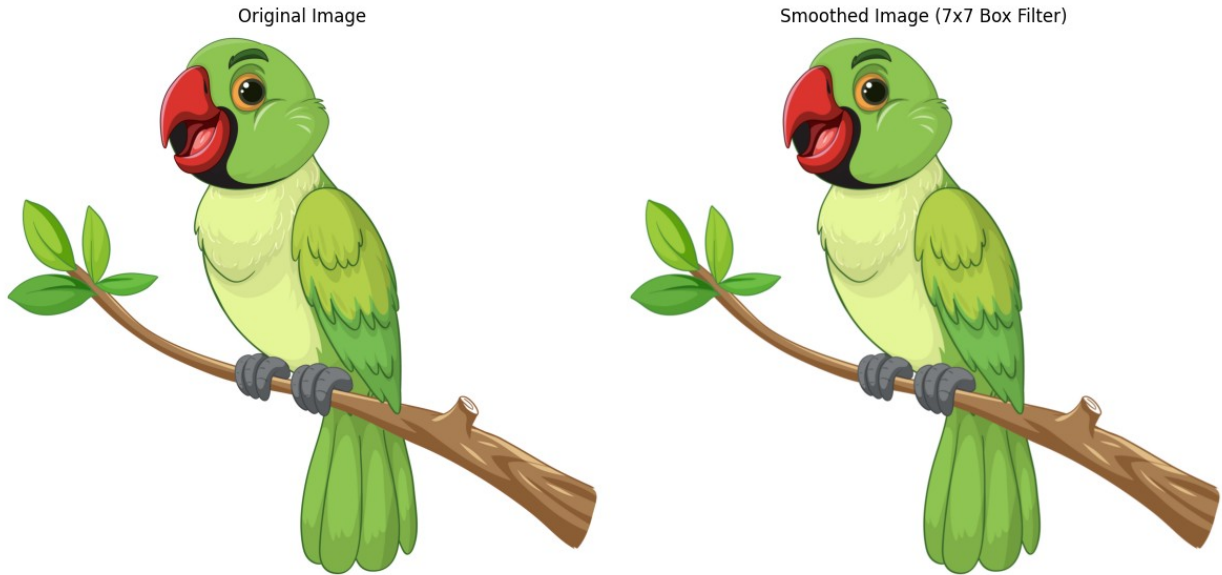
smoothed_box = cv2.blur(image, (7, 7))

fig, ax = plt.subplots(1, 2, figsize=(12, 6))

ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(smoothed_box)
ax[1].set_title("Smoothed Image (7x7 Box Filter)")
ax[1].axis("off")

plt.tight_layout()
plt.show()
```



Task#2

```
image = cv2.imread('Parrot.jpg')
gaussian_5x5 = cv2.GaussianBlur(image, (5, 5), 0)
gaussian_21x21 = cv2.GaussianBlur(image, (21, 21), 0)

fig, ax = plt.subplots(1, 3, figsize=(18, 6))

ax[0].imshow(image)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(gaussian_5x5)
ax[1].set_title("5x5 Gaussian Filter")
ax[1].axis("off")

ax[2].imshow(gaussian_21x21)
ax[2].set_title("21x21 Gaussian Filter")
ax[2].axis("off")

plt.tight_layout()
plt.show()
```



Task#3

```
image = cv2.imread('Parrot.jpg')

image_float = image.astype(np.float32) / 255.0

laplacian = cv2.Laplacian(image_float, ddepth=cv2.CV_32F, ksize=3)
sharpened = np.clip(image_float - laplacian, 0, 1)

fig, ax = plt.subplots(1, 3, figsize=(18, 6))

ax[0].imshow(image_float)
ax[0].set_title("Original Image")
ax[0].axis("off")

ax[1].imshow(np.abs(laplacian), cmap='gray' if len(image.shape) == 2
else None)
ax[1].set_title("Laplacian Filter Result")
ax[1].axis("off")

ax[2].imshow(sharpened)
ax[2].set_title("Sharpened Image")
ax[2].axis("off")

plt.tight_layout()
plt.show()
```

Clipping input data to the valid range for imshow with RGB data ([0..1] for floats or [0..255] for integers). Got range [0.0..6.454902].

Original Image



Laplacian Filter Result



Sharpened Image

